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Reflection at the free surface of the orthotropic piezo-hygro-thermo-elastic medium

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Abstract

Purpose – This paper aims to study the impact of pyro-electricity, moisture and temperature diffusivity on the energy distribution of plane waves at the free surface of an orthotropic piezo-hygro-thermo-elastic medium.

Design/methodology/approach – This study presents the novel creation of governing equations for an anisotropic piezothermoelastic medium with moisture impact, which is a significant contribution of this paper.

Findings – In addition to providing numerical data for the amplitude ratios and energy ratios of reflected waves, this study identifies five different kinds of coupled reflected plane waves, namely, quasi-longitudinal P wave, quasi-thermal wave, quasi-transverse wave, quasi-moisture wave and electric potential wave.

Research limitations/implications – The graphical analysis examines the impact of various factors, such as the angle of incidence, moisture and temperature diffusivity, pyro-electricity and frequency, on energy distribution.

Practical implications – This paper's results significantly impact the development of more efficient piezoelectric materials and their applications in geophysics.

Originality/value – The authors of the submitted document initiated and produced it collectively, with equal contributions from all members.

Keywords Piezoelectric, Pyro-electricity, Moisture, Hygro-thermal, Reflection, Amplitude ratios, Energy ratios

Paper type Research paper

Nomenclature

Θ = Thermal temperature;
c_{ijkl} = Elastic stiffness tensor;
ρ = Density;
ϕ = Electrical potential;
E_i = Electric field density;
τ_i, d_i = Pyro-electric constants;
J = Moisture concentration potential;

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q_{ij}^{m} = Coupled moisture diffusivities;
a_{ij}^{T} = Coupling thermal material coefficients;
D_{ij}^{T} = Temperature diffusivity;
β_{ij}^{m} = Moisture moduli tensor;
S = Entropy;
e_{ij} = Component of strain;
β_{ij}^{T} = Thermal moduli tensor;
D_{i} = Electric displacement;
η_{ijr},ε_{ij} = Piezo-thermal moduli tensors;
σ_{ij} = Components of the stress;
C_{E} = Specific heat at constant strain;
K_{m} = Moisture diffusion constant;
Q_{ij}^{T} = Coupled thermal diffusivity;
v = Poisson's ratio;
D_{ij}^{m} = Moisture diffusivity;
G = Shear modulus; and
δ_{ij} = Kroneckar delta.

1. Introduction

Most modern engineering materials, such as bio tissues, concrete, composites and foams, are heat sensitive (thermoscopic) and hygroscopic because of the material's porosity, crack and atomic structure. Smart materials, especially piezoelectric materials, can combine mechanical, electrical and hygro-thermal properties to improve the performance of networks through active stiffening, active damping and reducing the size of vibrations (Smith, 2005). Heat and moisture cannot be disregarded when addressing the mechanics of these materials and structures (Sih *et al.*, 1986; Hens, 2012). "Hygro-thermal effect" refers to the interaction between temperature and moisture. In light of this, that is crucial to continue researching the piezoelectric composite materials behave in a hygro-thermal environment.

Piezoelectric materials have garnered much attention in the domains of actuation, sensing, intelligent structures and mechanical systems. The terminology "piezoelectricity" describes a crystal's capacity to produce a voltage when stressed mechanically. Piezoelectricity was initially discovered by the Curie brothers (Curie and Curie, 1880). The governing equation for coupling between the temperature and piezoelectric material (piezo-thermo-elastic material) has been developed by Mindlin (1974). The fundamental physical laws for piezo-thermo-elastic materials have been investigated by Nowacki (1978, 1979). Researchers and engineers put a lot of effort into perfect piezoelectric materials in the past few decades and made their commercial use a reality. Lewis and his team investigated different heat and fluid flow issue models, as described in references (Lewis and Garner, 1972; Lewis *et al.*, 1991, 1997, 2004). Marin *et al.* (2013) conducted a study on the theory of thermoelasticity, which incorporates heat conduction in deformable bodies and accounts for the influence of two temperatures. Alzahrani *et al.* (2020) used the eigenvalues method to investigate a two-dimensional porous material with varying conductivity, including weak, strong and normal conductivity. Numerous researchers have investigated various problems related to the reflection and transmission of plane waves (Sharma and Marin, 2013; Marin *et al.*, 2017; Yadav, 2020, 2022a; Pathania *et al.*, 2022; Yadav, 2022b; Pathania *et al.*, 2023).

Raja *et al.* (2004b), using the virtual work concept, demonstrate the impact of active stiffness on the dynamic behavior of piezo-hygro-thermo-elastic (PHTE) laminates and a coupled piezoelectric finite element formulation with a hygro-thermal strain field has been

developed. PHTE curved and flat plate actuation and sensing behavior with active control were demonstrated by Raja *et al.* (2004a), using the first-order shear deformation theory and hygro-thermal strain. Based on PHTE differential equations, Wang *et al.* (2005) investigated the moisture impact on the response history and distribution of dynamical stresses in piezoelectric actuator layers during free vibration. The temperature, moisture and other non-linear effects on the mathematical modeling of piezoelectric intelligent structures using different vibration control approach and relevant software have been reviewed by Gupta *et al.* (2011).

Dini and Abolbashari (2016) investigated the thick-walled cylinder formed of functionally graded piezo smart materials in the context of a non-axisymmetric PHTE mechanical stress. The general steady-state solution for three-dimensional PHTE uses the potential theory technique from Zhao *et al.* (2018). Zenkour and Alghanmi (2021) analyzed the stress and deformation of sandwich plates with a functionally graded core and piezoelectric material faces with electro-mechanical-hygro-thermo sinusoidal loadings. The energy ratio at various types of interfaces, including fluid, elastic, semiconductor and piezo-thermo-elastic, has been studied by Yadav (2021a, 2021b, 2021c), Gupta and his team (Gupta *et al.*, 2023b; Barak *et al.*, 2023a; Gupta *et al.*, 2023a; Barak *et al.*, 2023b; Barak and Gupta, 2023; Gupta and Barak, 2023) and analyzed the effects of moisture, magnetic fields, phase lags, memory-dependent and fractional derivatives.

Based on the literature, piezoelectric materials are extensively used in a variety of real-world applications. Still, these materials' behavior at the orthotropic PHTE medium's free surface is not fully understood. The objective is to create a new mathematical model for such a medium, solve the reflection problem using the plane wave method and examine the effects of moisture and temperature diffusivities, moisture content, pyro-electricity and frequency on the distribution of energy ratio.

2. Basic equations of piezo-hygro-thermo-elastic medium

Following Szekeres (2000, 2012), Fourier's heat flux q_i^F depends upon the temperature diffusivity $\left(D_{ij}^T\right)$ and can be expressed as $q_i^F = -D_{ij}^T \Theta_{,i}$. In Fick's law, the moisture flux potential (f^F) depends upon the moisture diffusivity $\left(D_{ij}^m\right)$ and can be expressed as $f_i^F = -D_{ij}^m m_{,i}$. But numerous experimental results (Sih *et al.*, 1986) show that heat and moisture impact the temperature field and the moisture content. The Dufour and Soret effects are representative of the coupling of heat and moisture. The Dufour effect involves the transport of heat because of non-uniform moisture concentration. In contrast, the Soret effect describes the flow of moisture arising from a non-uniform temperature within a solid material with a uniform moisture distribution. Because of Dufour flux (q^d) and Soret flux (f^s), cross-coupling diffusivities q_{ij}^m, Q_{ij}^T come into existence such that $q_i^d = -q_{ij}^m m_{,i} f_i^s = -Q_{ij}^T \Theta_{,j}$.

Following the works of Gupta *et al.* (2023b), Szekeres (2000, 2012), Hosseini and Ghadiri Rad (2017) and Yadav *et al.* (2022), the total heat flux q_i, total moisture flux f_i and the fundamental relations for a coupled anisotropic PHTE medium are:

$$q_i = -D_{ij}^T \Theta_{,i} - q_{ij}^m m_{,i}, \tag{2.1}$$

$$f_i = -D_{ij}^m m_{,,i} - Q_{ij}^T \Theta_{,i}. \tag{2.2}$$

The piezo-hygro-electric medium facilitates the propagation of various types of waves, including thermal, elastic, moisture and electromagnetic waves. The constitutive equations are given as:

$$D_i = \eta_{ijk} e_{jk} + \varepsilon_{ij} E_j + \tau_i \Theta + d_i m, E_i = -\phi_{,_i}, \tag{2.3}$$

$$\rho \Theta_0 \dot{S} = -q_{i,j}, \rho \Theta_0 S = \Theta + \frac{\beta_{ij}^T \Theta_0}{\rho C_E} e_{kk} - \Theta_0 \tau_i \phi_{,i}, \tag{2.4}$$

$$-q_{i,j} = D_{ij}^T \Theta_{,ij} + q_{ij}^m m_{,ij}, \tag{2.5}$$

$$\left.\begin{aligned} \dot{J} &= -f_{i,j}, \\ J &= m + \frac{\beta_{ij}^m D_{ij}^m m_0}{K_m} e_{kk} - m_0 d_i \phi_{,i} \end{aligned}\right\}, \tag{2.6}$$

$$-f_{i,j} = D_{ij}^m m_{,ij} + Q_{ij}^T \Theta_{,ij}, \tag{2.7}$$

$$\sigma_{ij} = c_{ijkl} e_{kl} - \eta_{ijk} E_k + \beta_{ij}^T \Theta + \beta_{ij}^m m, \tag{2.8}$$

$$\text{where} \quad c_{ijkl} = \frac{2Gv}{1-2v} \delta_{ij} \delta_{kl} + G \delta_{ik} \delta_{jl} + G \delta_{il} \delta_{jk}, \quad i,j,k,l = 1,2,3.$$

Now, introduce the compact notation by replacing the pairs of indices ij or kl with single indices p or q and taking the values 1, 2, 3, 4, 5 and 6 according to:

ij or kl:	11	22	33	23 or 32	31 or 13	12 or 21
p or q:	1	2	3	4	5	6

Thus, $c_{ijkl} \rightarrow c_{pq}$, $\eta_{ikl} \rightarrow \eta_{ip}$:

$$\left.\begin{aligned} \sigma_{11} &= c_{11} u_{1,1} + c_{12} u_{2,2} + c_{13} u_{3,3} - \beta_1^T \Theta - \beta_1^m m + \eta_{31} \phi_{,3}, \\ \sigma_{12} &= c_{66} (u_{1,2} + u_{2,1}), \\ \sigma_{13} &= c_{55} (u_{1,3} + u_{3,1}) + \eta_{15} \phi_{,1}, \\ \sigma_{22} &= c_{12} u_{1,1} + c_{22} u_{2,2} + c_{23} u_{3,3} - \beta_2^T \Theta - \beta_2^m m + \eta_{32} \phi_{,3}, \\ \sigma_{23} &= c_{44} (u_{2,3} + u_{3,2}) + \eta_{24} \phi_{,2}, \\ \sigma_{33} &= c_{13} u_{1,1} + c_{23} u_{2,2} + c_{33} u_{3,3} - \beta_3^T \Theta - \beta_3^m m + \eta_{33} \phi_{,3}, \end{aligned}\right\} \tag{2.9}$$

$$\left.\begin{aligned} D_1 &= \eta_{15} (u_{1,3} + u_{3,1}) - \varepsilon_{11} \phi_{,1}, \\ D_2 &= \eta_{24} (u_{2,3} + u_{3,2}) - \varepsilon_{22} \phi_{,2} \\ D_3 &= \eta_{31} u_{1,1} + \eta_{32} u_{2,2} + \eta_{33} u_{3,3} - \varepsilon_{33} \phi_{,3} + \tau_3 \Theta + d_3 m \end{aligned}\right\}. \tag{2.10}$$

Equation of motion:

$$\sigma_{ji,j} = \rho\ddot{u}_i. \tag{2.11}$$

Using equations (2.1), (2.4) and (2.5), we get:

$$D_i^T \Theta_{,ii} + q_i^m m_{,ii} - \dot{\Theta} - \frac{\beta_i^T \Theta_0}{\rho C_E}\dot{u}_{j,j} + \Theta_0 \tau_i \dot{\phi}_{,i} = 0. \tag{2.12}$$

Using equations (2.2), (2.6) and (2.7), we get:

$$D_i^m m_{,ii} + Q_i^T \Theta_{,it} - \dot{m} - \frac{\beta_i^m D_i^m m_0}{K_m}\dot{u}_{j,j} + m_0 d_i \dot{\phi}_{,i} = 0. \tag{2.13}$$

The electric displacement equation (2.3) can be expressed as:

$$D_{i,i} = 0 \Rightarrow D_{1,1} + D_{2,2} + D_{3,3} = 0, \tag{2.14}$$

where $\beta_{ij}^T = \beta_i^T \delta_{ij}$, $\beta_{ij}^m = \beta_i^m \delta_{ij}$, $q_{ij}^m = q_i^m \delta_{ij}$, $Q_{ij}^T = Q_i^T \delta_{ij}$, $D_{ij}^T = D_i^T \delta_{ij}$, $D_{ij}^m = D_i^m \delta_{ij}$, and, i, j is not summed.

3. Problem formulation

We consider a homogeneous orthotropic PHTE half-space with the Cartesian coordinate system (x, y, z) in uniform initial moisture m_0 and temperature $\boldsymbol{\Theta}_0$ and designate such that the $y = 0$. The hygro-thermo-elastic plane waves propagate in the xz plane such that all field variables depend on x, z, t and the displacement vector is represented as $\overrightarrow{u} = (u, 0, w)$. The governing equation (2.9)–(2.14) in the xz plane can be written as:

$$c_{11}\frac{\partial^2 u}{\partial x^2} + c_{55}\frac{\partial^2 u}{\partial z^2} + (c_{13} + c_{55})\frac{\partial^2 w}{\partial x \partial z} + (\eta_{31} + \eta_{15})\frac{\partial^2 \phi}{\partial x \partial z} - \beta_1^T \frac{\partial \Theta}{\partial x} - \beta_1^m \frac{\partial m}{\partial x} = \rho\ddot{u}, \tag{3.1}$$

$$(c_{13} + c_{55})\frac{\partial^2 u}{\partial x \partial z} + c_{55}\frac{\partial^2 w}{\partial x^2} + c_{33}\frac{\partial^2 w}{\partial z^2} + \eta_{15}\frac{\partial^2 \phi}{\partial x^2} + \eta_{33}\frac{\partial^2 \phi}{\partial z^2} - \beta_3^T \frac{\partial \Theta}{\partial z} - \beta_3^m \frac{\partial m}{\partial z} = \rho\ddot{w}, \tag{3.2}$$

$$D_1^T \frac{\partial^2 \Theta}{\partial x^2} + D_3^T \frac{\partial^2 \Theta}{\partial z^2} + q_1^m \frac{\partial^2 m}{\partial x^2} + q_3^m \frac{\partial^2 m}{\partial z^2} - \frac{\partial \Theta}{\partial t} - \frac{\Theta_0}{\rho C_E}\left(\beta_1^T \frac{\partial^2 u}{\partial x \partial t} + \beta_3^T \frac{\partial^2 w}{\partial z \partial t}\right) + \Theta_0 \tau_3 \frac{\partial^2 \phi}{\partial z \partial t} = 0, \tag{3.3}$$

$$Q_1^T \frac{\partial^2 \Theta}{\partial x^2} + Q_3^T \frac{\partial^2 \Theta}{\partial z^2} + D_1^m \frac{\partial^2 m}{\partial x^2} + D_3^m \frac{\partial^2 m}{\partial z^2} - \frac{\partial m}{\partial t} - \frac{m_0}{K_m}\left(\beta_1^T D_1^m \frac{\partial^2 u}{\partial x \partial t} + \beta_3^T D_3^m \frac{\partial^2 w}{\partial z \partial t}\right) + m_0 d_3 \frac{\partial^2 \phi}{\partial z \partial t} = 0, \tag{3.4}$$

$$(\eta_{15} + \eta_{31})\frac{\partial^2 u}{\partial x \partial z} + \eta_{15}\frac{\partial^2 w}{\partial x^2} + \eta_{33}\frac{\partial^2 w}{\partial z^2} - \varepsilon_{11}\frac{\partial^2 \phi}{\partial x^2} - \varepsilon_{33}\frac{\partial^2 \phi}{\partial z^2} + \tau_3 \frac{\partial \Theta}{\partial z} + d_3 \frac{\partial m}{\partial z} = 0. \tag{3.5}$$

where $\beta_1^T = (c_{11} + c_{12})\alpha_{11}^T + c_{13}\alpha_{33}^T$, $\beta_3^T = 2c_{13}\alpha_{11}^T + c_{33}\alpha_{33}^T$, $\beta_1^m = (c_{11} + c_{12})\alpha_{11}^m + c_{13}\alpha_{33}^m$ and $\beta_3^m = 2c_{13}\alpha_{11}^m + c_{33}\alpha_{33}^m$, $\alpha_{11}^T, \alpha_{33}^T$ are the constants of linear thermal expansion, and $\alpha_{11}^m, \alpha_{33}^m$ are the constants of moisture expansion.

4. Plane-wave solution and reflection from the free surface

Let us assume the plane wave solution of the form:

$$(u, w, \phi, \Theta, m) = \left(u', w', \phi', \Theta', m'\right)(z)e^{\iota(\omega t - kx)}, \tag{4.1}$$

where u', w', ϕ', Θ' and m' are the function of z only; k and ω are the wave number and angular frequency, respectively.

Equations (3.1)–(3.5) with the aid of equation (4.1) yield a homogenous system of equations in ω, k and five unknowns u', w', ϕ', Θ' and m' as:

$$\Omega R = 0, \tag{4.2}$$

where:

$$\Omega = \begin{bmatrix} a_{11} + c_{55}D^2 & -a_{12}D & -a_{13}D & a_{14} & a_{15} \\ a_{21}D & a_{22} - c_{33}D^2 & a_{23} - \eta_{33}D^2 & \beta_3^T D & \beta_3^m D \\ a_{31} & a_{32}D & -a_{33}D & a_{34} - D_3^T D^2 & a_{35} - q_3^m D^2 \\ a_{41} & a_{42}D & -a_{43}D & a_{44} - Q_3^T D^2 & a_{45} - D_3^m D^2 \\ a_{51}D & a_{52} - \eta_{33}D^2 & \varepsilon_{33}D^2 - a_{53} & -\tau_3 D & -d_3 D \end{bmatrix} R = \begin{bmatrix} u' \\ w' \\ \phi' \\ \Theta' \\ m' \end{bmatrix},$$

and $a_{11} = \rho\omega^2 - c_{11}k^2$, $a_{12} = (c_{13} + c_{55})\iota k$, $a_{13} = (\eta_{31} + \eta_{15})\iota k$, $a_{14} = \beta_1^T \iota k$, $a_{15} = \beta_1^m \iota k$, $D = \partial/\partial z$, $a_{21} = (c_{13} + c_{55})\iota k$, $a_{22} = c_{55}k^2 - \rho\omega^2$, $a_{23} = \eta_{15}k^2$, $a_{31} = \frac{\Theta_0 \beta_1^T k\omega}{\rho C_e}$, $a_{32} = \frac{\Theta_o \beta_3^T \iota\omega}{\rho C_e}$, $a_{33} = T_0\tau_3\iota\omega$, $a_{34} = k^2 D_1^T + \iota\omega$, $a_{35} = k^2 q_1^m$, $a_{41} = \frac{m_0 \beta_1^T D_1^m k\omega}{K_m}$, $a_{42} = \frac{m_0 \beta_3^T D_3^m \iota\omega}{K_m}$, $a_{43} = m_0 d_3 \iota\omega$, $a_{44} = Q_1^T k^2$, $a_{45} = k^2 D_1^m + \iota\omega$, $a_{51} = (\eta_{15} + \eta_{31})\iota k$, $a_{52} = \eta_{15}k^2$, $a_{53} = \varepsilon_{11}k^2$.

The coefficient matrix determinant in equation (4.2) must be null to obtain a solution that is not trivial, which results in the following characteristics equation:

$$\left(B_0 D^{10} + B_1 D^8 + B_2 D^6 + B_3 D^4 + B_4 D^2 + B_5\right)\left\{u'(z), w'(z), \phi'(z), m'(z), \Theta'(z)\right\} = 0, \tag{4.3}$$

The coefficients of this equation are written in Appendix.

Equation (4.3) can be factored as:

$$\left(D^2 - \lambda_1^2\right)\left(D^2 - \lambda_2^2\right)\left(D^2 - \lambda_3^2\right)\left(D^2 - \lambda_4^2\right)\left(D^2 - \lambda_5^2\right)\left\{u'(z), w'(z), \phi'(z), m'(z), \Theta'(z)\right\} = 0. \tag{4.4}$$

With the help of MATLAB software, the roots of equation (4.3) were determined and arranged in decreasing order. The roots λ_1, λ_2, λ_3, λ_4 and λ_5 represent the velocities of five reflected plane waves, that is, quasi-longitudinal P (QP) wave, quasi-thermal (QT) wave, quasi-transverse (QS) wave, quasi-moisture (QM) wave and electric potential (QEP) wave, respectively.

5. Reflection coefficient

A plane wave making an angle θ_0 with the z-axis, incident at the free surface of homogenous orthotropic PHTE half-space. This results in five waves reflected, as shown in Figure 1.

The formal solutions for the mechanical displacements, electric potential, temperature and moisture in the considered medium are:

$$u'(z) = \sum_{i=1}^{5} A_{0i} e^{-\lambda_i z} + \sum_{i=1}^{5} A_i e^{\lambda_i z}, \tag{5.1}$$

$$w'(z) = \sum_{i=1}^{5} p_i A_{0i} e^{-\lambda_i z} + \sum_{i=1}^{5} p_i A_i e^{\lambda_i z}, \tag{5.2}$$

$$\phi'(z) = \sum_{i=1}^{5} q_i A_{0i} e^{-\lambda_i z} + \sum_{i=1}^{5} q_i A_i e^{\lambda_i z}, \tag{5.3}$$

$$\Theta'(z) = \sum_{i=1}^{5} r_i A_{0i} e^{-\lambda_i z} + \sum_{i=1}^{5} r_i A_i e^{\lambda_i z}, \tag{5.4}$$

$$m'(z) = \sum_{i=1}^{5} s_i A_{0i} e^{-\lambda_i z} + \sum_{i=1}^{5} s_i A_i e^{\lambda_i z} \tag{5.5}$$

where:

$$\begin{bmatrix} p_i \\ q_i \\ r_i \\ s_i \end{bmatrix} = \begin{bmatrix} a_{12}\lambda_i & a_{13}\lambda_i & a_{14} & a_{15} \\ a_{22} - c_{33}\lambda_i^2 & a_{23} - \eta_{33}\lambda_i^2 & -\beta_3^T \lambda_i & -\beta_3^m \lambda_i \\ a_{32}\lambda_i & -a_{33}\lambda_i & -a_{34} + D_3^T \lambda_i^2 & -a_{35} + q_3^m \lambda_i^2 \\ a_{42}\lambda_i & -a_{43}\lambda_i & -a_{44} + Q_3^T \lambda_i^2 & -a_{45} + D_3^m \lambda_i^2 \end{bmatrix}^{-1} \begin{bmatrix} -a_{11} - c_{55}\lambda_i^2 \\ \lambda_i a_{21} \\ a_{31} \\ a_{41} \end{bmatrix}.$$

6. Boundary conditions

Assume the following appropriate boundary conditions at $z = 0$ as:

$$\sigma_{33} = 0,\ \sigma_{31} = 0,\ \frac{\partial \Theta}{\partial z} = 0,\ \frac{\partial m}{\partial z} = 0,\ D_3 = 0 \tag{6.1}$$

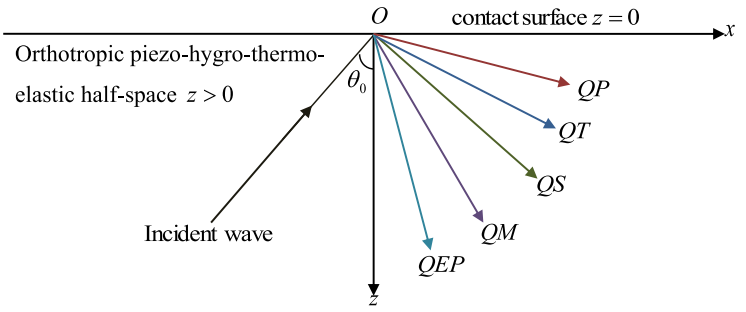


Figure 1.
Geometry of the problem

Using equations (5.1)–(5.5), we can see that the boundary requirements are met if and only if:

$$\frac{\sin\theta_0}{v_0} = \frac{k}{\omega}, \tag{6.2}$$

where v_0 is the incident wave velocity.

Equations (4.1) and (5.1)–(5.5) are used in boundary conditions (6.1) with the help of equation (6.2) to produce a system of five nonhomogeneous equations as:

$$\begin{bmatrix} \Delta_{11} & \Delta_{12} & \Delta_{13} & \Delta_{14} & \Delta_{15} \\ \Delta_{21} & \Delta_{22} & \Delta_{23} & \Delta_{24} & \Delta_{25} \\ \Delta_{31} & \Delta_{32} & \Delta_{33} & \Delta_{34} & \Delta_{35} \\ \Delta_{41} & \Delta_{42} & \Delta_{43} & \Delta_{44} & \Delta_{45} \\ \Delta_{51} & \Delta_{52} & \Delta_{53} & \Delta_{54} & \Delta_{55} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \end{bmatrix}, \tag{6.3}$$

where:

$$X_i = \frac{A_i}{A^*}; i = 1-5$$

For incidence QP wave: $A^* = A_{01}$, $Q_i = \Delta^0_{i1}; i = 1-5$.

For incidence QT wave: $A^* = A_{02}$, $Q_i = \Delta^0_{i2}; i = 1-5$.

For incidence QS wave: $A^* = A_{03}$, $Q_i = \Delta^0_{i3}; i = 1-5$.

For incidence QM wave: $A^* = A_{04}$, $Q_i = \Delta^0_{i4}; i = 1-5$.

For incidence QEP wave: $A^* = A_{05}$, $Q_i = \Delta^0_{i5}; i = 1-5$.

$$\Delta_{1j} = \left(-c_{13}\imath k + c_{33}\lambda_j p_j + \eta_{33}\lambda_j q_j - \beta_3^T r_j - \beta_3^m s_j\right); j = 1-5$$

$$\Delta^0_{1j} = \left(c_{13}\imath k + c_{33}\lambda_j p_j + \eta_{33}\lambda_j q_j + \beta_3^T r_j + \beta_3^m s_j\right); j = 1-5$$

$$\Delta_{2j} = \left\{c_{55}\left(\lambda_j - \imath k p_j\right) - \eta_{15} q_j \imath k\right\}; j = 1-5,$$

$$\Delta^0_{2j} = \left\{c_{55}\left(\lambda_j + \imath k p_j\right) + \eta_{15} q_j \imath k\right\}; j = 1-5$$

$$\Delta_{3j} = r_j\lambda_j; j = 1-5,$$

$$\Delta^0_{3j} = r_j\lambda_j; j = 1-5,$$

$$\Delta_{4j} = s_j\lambda_j; j = 1-5$$

$$\Delta_{4j}^{0} = s_j\lambda_j; j = 1-5$$

$$\Delta_{5j} = \left(-\eta_{31}\iota k + \eta_{33}\lambda_j p_j - \varepsilon_{33}\lambda_j q_j + \tau_3 r_j + d_3 s_j\right); j = 1-5$$

$$\Delta_{5j}^{0} = \left(\eta_{31}\iota k + \eta_{33}\lambda_j p_j - \varepsilon_{33}\lambda_j q_j - \tau_3 r_j - d_3 s_j\right); j = 1-5$$

7. Energy ratios

Consider the energy distribution among several reflected waves at a surface element with a unit area. For an orthotropic PHTE material, following Achenbach (1975), the normal acoustic flux at a free surface $z = 0$ is given by:

$$P = -\mathrm{Re}\left(\sigma_{31}\overline{\dot{u}} + \sigma_{33}\overline{\dot{w}} - \phi\overline{\dot{D}}_3 + D_3^T\frac{\Theta}{\Theta_0}\frac{\partial\overline{\Theta}}{\partial z} + D_3^m\frac{m}{m_0}\frac{\partial\overline{m}}{\partial z}\right), \tag{7.1}$$

where Re for the real portion, the above bar represents the complex conjugate.

The average energy of incident waves at $z = 0$:

$$< P_{inc,i} >= -\mathrm{Re}\frac{\iota\omega}{2}\left(\bar{p}_i\Delta_{1i}^{0} + \Delta_{2i}^{0} - q_i\bar{\Delta}_{5i}^{0} - \frac{D_3^T}{\iota\omega\Theta_0}r_i\bar{\Delta}_{3i}^{0} - \frac{D_3^m}{\iota\omega m_0}s_i\bar{\Delta}_{4i}^{0}\right)|A_{0i}|^2; i = 1-5, \tag{7.2}$$

$$\text{where}\begin{cases} i = 1, \text{ for incident } QP \text{ wave} \\ i = 2, \text{ for incident } QT \text{ wave} \\ i = 3, \text{ for incident } QS \text{ wave} \\ i = 4, \text{ for incident } QM \text{ wave} \\ i = 5, \text{ for incident } QEP \text{ wave} \end{cases}.$$

The average energy of reflected waves at $z = 0$:

$$< P_{ref,i} >= \mathrm{Re}\frac{\iota\omega}{2}\left(\bar{p}_i\Delta_{1i} + \Delta_{2i} - q_i\bar{\Delta}_{5i} - \frac{D_3^T}{\iota\omega\Theta_o}r_i\bar{\Delta}_{3i} - \frac{D_3^m}{\iota\omega m_o}s_i\bar{\Delta}_{4i}\right)|A_i|^2; i = 1-5. \tag{7.3}$$

The energy ratios of:

$$\text{reflected waves}; E_i = \frac{< P_{ref,i} >}{< P_{inc,i} >}, i = 1-5, \tag{7.4}$$

$$\text{interaction waves; } E_{\text{int}} = \sum_{m,n=1,m\neq n}^{5} E_{mn}, \text{ where } E_{mn} = \frac{< P_{mn} >}{< P_{inc,i} >}, i = 1-5, \tag{7.5}$$

and $< P_{mn} >= \text{Re}\,\frac{i\omega}{2}\left(\overline{p}_n\Delta_{1m}A_m\overline{A}_n + \Delta_{2m}A_m\overline{A}_n - q_n\overline{\Delta}_{5m}\overline{A}_mA_n - \frac{D_3^T}{l\omega T_o}r_n\overline{\Delta}_{3m}\overline{A}_mA_n - \frac{D_3^m}{l\omega m_o}s_n\overline{\Delta}_{4m}\,\overline{A}_mA_n\right)$. The energy will be conserved if $\sum_{i=1}^{5} E_i + E_{\text{int}} = 1$.

8. Special case (hygro-thermo-elastic case)

Neglecting orthotropy, piezoelectric and pyro-electric effects, we have: $c_{11} = \lambda + 2\mu$, $c_{33} = \lambda + 2\mu$, $c_{13} = \lambda$, $c_{55} = \mu$, $\beta_1^T = \beta_3^T = \beta^T$, $\beta_1^m = \beta_3^m = \beta^m$, $D_1^T = D_3^T = D^T$, $D_1^m = D_3^m = D^m$, $q_1^m = q_3^m = D_T^m$, $Q_1^T = Q_3^T = D_m^T$, $d_3 = 0$, $\tau_3 = 0$, $\varepsilon_{11} = \varepsilon_{33} = 0$, $\eta_{31} = \eta_{33} = \eta_{15} = 0$, the above analysis reduces to hygro-thermo-elastic model studied by Yadav *et al.* (2022).

Equations (3.1)–(3.5) reduce to the following equations:

$$(\lambda + 2\mu)\frac{\partial^2 u}{\partial x^2} + \mu\frac{\partial^2 u}{\partial z^2} + ((\lambda + \mu))\frac{\partial^2 w}{\partial x \partial z} + -\beta^T\frac{\partial \Theta}{\partial x} - \beta^m\frac{\partial m}{\partial x} = \rho\ddot{u}, \tag{8.1}$$

$$(\lambda + \mu)\frac{\partial^2 u}{\partial x \partial z} + \mu\frac{\partial^2 w}{\partial x^2} + (\lambda + 2\mu)\frac{\partial^2 w}{\partial z^2} + -\beta^T\frac{\partial \Theta}{\partial z} - \beta^m\frac{\partial m}{\partial z} = \rho\ddot{w}, \tag{8.2}$$

$$D^T\left(\frac{\partial^2 \Theta}{\partial x^2} + \frac{\partial^2 \Theta}{\partial z^2}\right) + D_T^m\left(\frac{\partial^2 m}{\partial x^2} + \frac{\partial^2 m}{\partial z^2}\right) - \frac{\partial \Theta}{\partial t} - \frac{\beta^T\Theta_0}{\rho C_E}\left(\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 w}{\partial z \partial t}\right) = 0, \tag{8.3}$$

$$D_m^T\left(\frac{\partial^2 \Theta}{\partial x^2} + \frac{\partial^2 \Theta}{\partial z^2}\right) + D^m\left(\frac{\partial^2 m}{\partial x^2} + \frac{\partial^2 m}{\partial z^2}\right) - \frac{\partial m}{\partial t} - \frac{\beta^m m_0 D^m}{K_m}\left(\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 w}{\partial z \partial t}\right) = 0. \tag{8.4}$$

9. Numerical results and discussion

Following Sih *et al.* (1986), Yadav *et al.* (2022) and Gupta *et al.* (2023b, 2023a), the cadmium selenide material constants and hygro parameters have been considered, as shown in Table 1.

Using MATLAB programming, equations (7.4) and (7.5) are computed for the CdSe material under hygro-thermal conditions. The energy ratios against the angle of incidence $0° < \theta_0 \leq 90°$, for incident plane waves at the interface $x_3 = 0$ are computed and plotted graphically for different values of moisture concentration m_0, moisture diffusivity D_1^m, D_3^m, temperature diffusivity D_1^T, D_3^T, pyro-electric parameter τ_3, d_3 and frequency ω. The energy ratios of reflected QP wave, QT wave, QS wave, QM wave and QEP wave, for incidence QP wave, are represented by E_{RP}, E_{RT}, E_{RS}, E_{RM} and E_{REP}, respectively. The E_{int} denotes the total interaction energy ratio between the reflected waves. The image's sub-figure depicts overlapping curves to illustrate the differences precisely and clearly, making it easier to understand the intricate details at the micro-scale.

Figures 2(a)–2(f) depicted the variations of the energy ratios with an angle of incidence $0° < \theta_0 \leq 90°$ for different values of moisture concentration $m_0 = 10\%, 20\%, 30\%$ and 40%. In Figures 2(a), 2(b) and 2(d), it is observed that as the angle of incidence increases, the

Symbol	Value
c_{11}	74.1×10^9 Nm^{-2}
c_{12}	45.2×10^9 Nm^{-2}
c_{13}	39.3×10^9 Nm^{-2}
c_{13}	83.6×10^9 Nm^{-2}
c_{55}	13.2×10^9 Nm^{-2}
Θ_0	298 K
β_1^T	6.21×10^5 Nm^{-2}K^{-1}
β_3^T	5.51×10^5 Nm^{-2}K^{-1}
m_0	10%
D_1^m	0.8×10^{-4} m^{2}sec^{-1}
D_3^m	1.1×10^{-4} m^{2}sec^{-1}
η_{33}	0.347 Cm^{-2}
q_1^m	18.5×10^{-5} m^2(^{0}K)/sec(%H_2O)
q_3^m	18×10^{-5} m^2(^{0}K)/sec(%H_2O)
K_m	2.2×10^{-8} Kg/msM
ε_{11}	8.26×10^{-11} C^{2}N^{-1}m^{-2}
ε_{33}	9.03×10^{-11} C^{2}N^{-1}m^{-2}
τ_3	-2.9×10^{-6} $Cm^{-2}K^{-1}$
β_1^m	0.8×10^{-4} Nm^{-2}K^{-1}
β_3^m	1.1×10^{-4} Nm^{-2}K^{-1}
C_E	260 Jkg^{-1}K^{-1}
P	5,504 kgm^{-3}
η_{31}	-0.160 Cm^{-2}
η_{15}	-0.138 Cm^{-2}
D_1^T	50 m^2 sec^{-1}
D_3^T	75 m^2 sec^{-1}
ω	750 Hz
Q_1^T	31.3×10^{-6} m^2(%H_2O)/sec(^{0}K)
Q_3^T	33.3×10^{-6} m^2(%H_2O)/sec(^{0}K)
d_3	0.34 Cm^{-2} K^{-1}

Source: Compiled by authors

Table 1. Values of the materials constants

magnitude of E_{RP}, E_{RT} and E_{RM} also increases except for the E_{RT}, which shows a minimum value for the $m_0 = 10\%$, at a certain angle $\theta = 35°$. However, as the moisture concentration increases, the magnitude of E_{RP}, E_{RT} and E_{RM} decreases. The moisture impact is negligible near the normal incidence, but it becomes significant as the angle of incidence increases, with the maximum impact occurring at the grazing incidence.

Figure 2(c) presents an intriguing finding: at normal incidence, the magnitude of E_{RS} experiences a steep decline as the angle of incidence is increased $0° \leq \theta \leq 20°$, and once the incidence angle reaches grazing incidence, the E_{RS} follows a nearly stationary path. Upon closer inspection of the magnified sub-indent picture confirms that the magnitude of E_{RS} significantly decreases with an increase in the amount of moisture. Moving on to Figure 2(e), we reveal that as the angle of incidence rises, the E_{REP} first goes to maxima at an angle $\theta = 3°$, then shows a minimum at $\theta = 19°$; after a further rise of angle $19° \leq \theta \leq 90°$, the magnitude of E_{REP} increases very slowly. Notably, an increase in the moisture content leads to a decrease in the magnitude of the $_{EREP}$ across the range of angles examined. In Figure 2(f), we can see that the interaction energy starts out as negative but becomes positive as the incidence angle is increased to $\theta = 53°$. The impact of moisture on the interaction energy at different angles of incidence is evident in the sub-indent image.

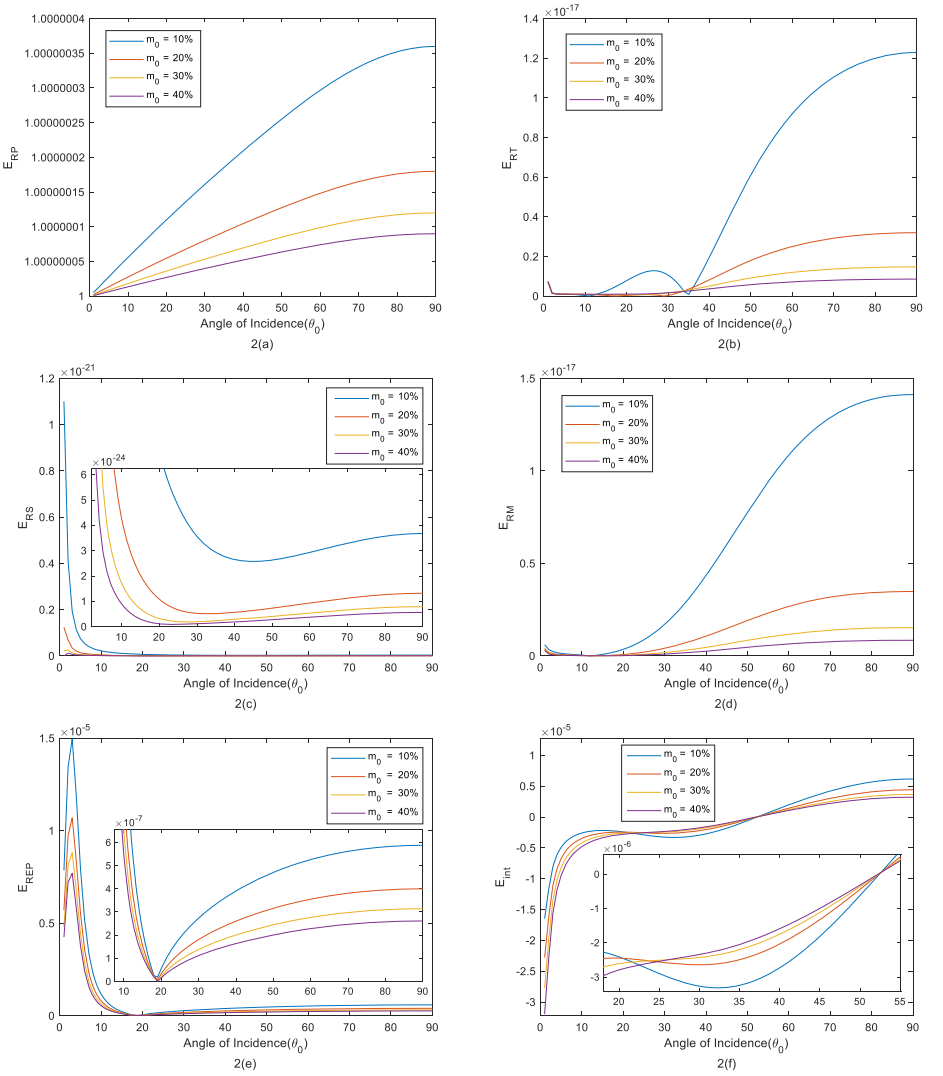
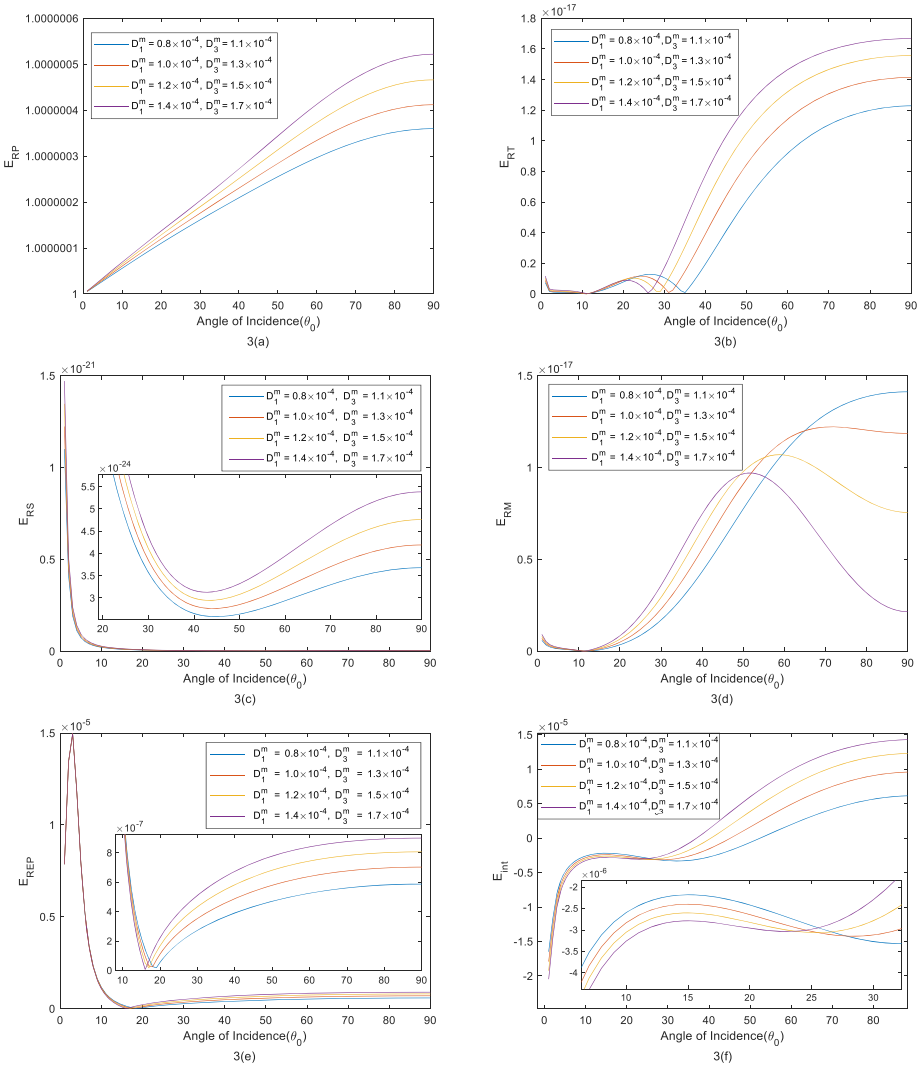


Figure 2.
Figures 2(a)–2(f). Variations of the energy ratios of reflected quasi-longitudinal P wave (E_{RP}), quasi-thermal wave (E_{RT}), quasi-transverse wave (E_{RS}), quasi-moisture wave (E_{RM}), electric potential wave (E_{REP}) and interaction energy (E_{int}) are depicted against the angle of incidence θ_0, $0 < \theta_0 < 90°$ for different values of moisture concentration $m_0 = 10\%, 20\%, 30\%$ and 40%, at when $D_1^T = 50, D_3^T = 75$, $D_1^m = 0.8 \times 10^{-4}$, $D_3^m = 1.1 \times 10^{-4}$, $\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$ and $\omega = 750$ Hz

Source: Created by author's

The variations of the energy ratios against the angle of incidence θ_0, $0 < \theta_0 \leq 90°$ for different values of moisture diffusivity parameters $\left(\mathrm{D}_1^{\mathrm{m}} = 0.8 \times 10^{-4}, D_3^m = 1.1 \times 10^{-4}\right)$, $\left(\mathrm{D}_1^{\mathrm{m}} = 1.0 \times 10^{-4}, D_3^m = 1.3 \times 10^{-4}\right)$, $\left(\mathrm{D}_1^{\mathrm{m}} = 1.2 \times 10^{-4}, D_3^m = 1.5 \times 10^{-4}\right)$ and $(\mathrm{D}_1^{\mathrm{m}} = 1.4 \times 10^{-4}, D_3^m = 1.7 \times 10^{-4})$ are shown in Figures 3(a)–3(f). The following observations are made:

- Increasing the moisture diffusivity parameters leads to an increase in the magnitudes of E_{RP}, E_{RS}. This increase in magnitude becomes more pronounced as the angle of incidence increases $0 < \theta_0 \leq 90°$.
- At first, the magnitude of E_{RT} is not significantly affected by the moisture diffusivity parameters. However, after a certain angle $\theta_0 > 29°$, the impact of moisture diffusivity parameters becomes noticeable.
- The magnitude of the E_{RM} in the range of angle of incidence $0 < \theta_0 \leq 50°$ increases with moisture diffusivity. After a further increase in the angle of incidence $50° < \theta_0 \leq 90°$, the effect of moisture diffusivity is inverted and becomes more substantial.



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Figure 3.
Figures 3(a)–3(f). Variations of the energy ratios of reflected quasi-longitudinal P wave (E_{RP}), quasi-thermal wave (E_{RT}), quasi-transverse wave (E_{RS}), quasi-moisture wave (E_{RM}), electric potential wave (E_{REP}) and interaction energy (E_{int}) are depicted against the angle of incidence θ_0, $0 < \theta_0 < 90°$ for different values of moisture diffusivity parameter $(D_1^{\text{m}} = 1.8 \times 10^{-4}, D_3^{\text{m}} = 1.1 \times 10^{-4})$, $(D_1^{\text{m}} = 1.0 \times 10^{-4}, D_3^{\text{m}} = 1.3 \times 10^{-4})$, $(D_1^{\text{m}} = 1.2 \times 10^{-4}, D_3^{\text{m}} = 1.5 \times 10^{-4})$ and $(D_1^{\text{m}} = 1.4 \times 10^{-4}, D_3^{\text{m}} = 1.7 \times 10^{-4})$, at when $m_0 = 10\%$, $D_1^T = 50, D_3^T = 75$, $D_1^m = 0.8 \times 10^{-4}$, $D_3^m = 1.1 \times 10^{-4}$, $\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$ and $\omega = 750$ Hz

- Near the normal incidence, the impact of moisture diffusivity parameters on magnitudes of E_{REP}, E_{int} is negligible. However, as the angle of incidence rises, the impact of moisture diffusivity parameters becomes more apparent and profound, particularly near the grazing incidence.

Figures 4(a)–4(f) display the variations of the energy ratios for different values of the temperature diffusivity parameter $\left(D_1^T = 50, D_3^T = 75\right)$, $\left(D_1^T = 100, D_3^T = 125\right)$, $\left(D_1^T = 150, D_3^T = 175\right)$ and $\left(D_1^T = 200, D_3^T = 225\right)$. By carefully analyzing Figures 4(a)–4(f), several key observations can be made:

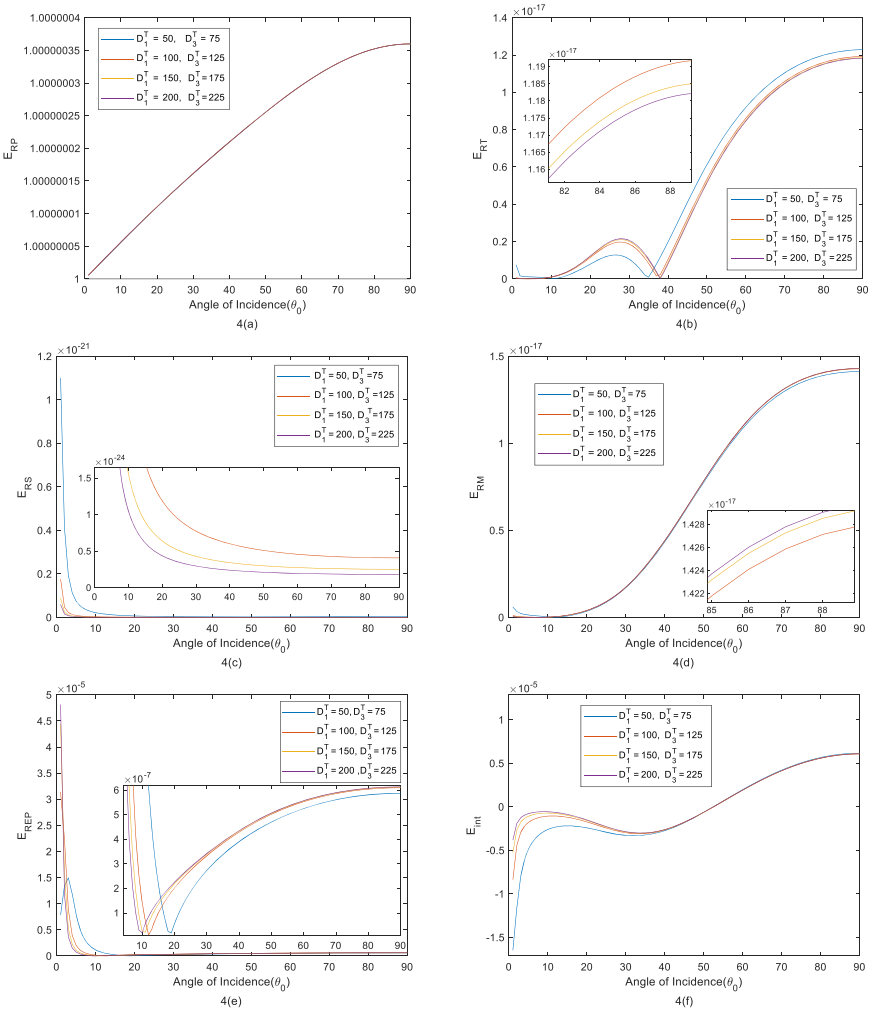


Figure 4.
Figures 4(a)–4(f). Variations of the energy ratios of reflected quasi-longitudinal P wave (E_{RP}), quasi-thermal wave (E_{RT}), quasi-transverse wave (E_{RS}), quasi-moisture wave (E_{RM}), electric potential wave (E_{REP}) and interaction energy (E_{int}) are depicted against the angle of incidence θ_0, $0 < \theta_0 < 90°$ for different values of temperature diffusivity parameter $(D_1^T = 50, D_3^T = 75)$, $(D_1^T = 100, D_3^T = 125)$, $(D_1^T = 150, D_3^T = 175)$, $(D_1^T = 200, D_3^T = 225)$, at when $m_0 = 10\%$, $D_1^m = 0.8 \times 10 - 4$, $D_3^m = 1.1 \times 10 - 4$, $\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$ and $\omega = 750$ Hz

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- It was found that no significant impact of temperature diffusivity has been observed in the magnitudes of E_{RP} and E_{RM} in the range of considered incidence angles.
- The magnitude of E_{RT} in the range of angle of incidence $0 < \theta_0 \leq 35°$ increases with temperature diffusivity. After a further increase in the angle of incidence $35° < \theta_0 \leq 90°$, the effect of temperature diffusivity is inverted and becomes more substantial.
- Increasing the temperature diffusivity parameters leads to a decrease in the magnitude of E_{RS} in $0 < \theta_0 \leq 90°$ and E_{REP} in $0 < \theta_0 \leq 15°$. The impact of temperature diffusivity on E_{RS} and E_{REP} is more prominent at lower angles of incidence and becomes progressively weaker at larger angles of incidence.
- The effect of temperature diffusivity parameters on magnitudes of E_{int} is significant at angles closer to normal incidence. However, as the angle of incidence rises $0 < \theta_0 \leq 90°$, the impact of temperature diffusivity parameters becomes less pronounced and eventually disappears.

Figures 5(a)–5(b) depict the effect of pyro-electric parameters on variations of the energy ratios for different values of pyroelectric parameters ($\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$) and ($\tau_3 = -3.04 \times 10^{-6}$ and $d_3 = 0.44$), ($\tau_3 = -3.14 \times 10^{-6}$ and $d_3 = 0.54$) and ($\tau_3 = -3.24 \times 10^{-6}$ and $d_3 = 0.64$). A thorough examination of Figures 5(a)–5(f) will reveal some significant findings that will help to better understand pyro-electricity and aid in making more educated choices:

- Increasing the pyro-electric parameters leads to a decrease in the magnitudes of E_{RP}, E_{RM}. This decrease in magnitude becomes more significant as the angle of incidence increases $0 < \theta_0 \leq 90°$.
- In the beginning, the magnitude of E_{RT} is not significantly affected by the pyro-electric parameters. However, after a certain angle $\theta_0 > 35°$, the influence of pyro-electric parameters becomes apparent.
- Upon closer examination of the magnified sub-indent picture confirms that the magnitudes of E_{RS}, E_{REP} decreases significantly as the pyro-electric parameters are increased. This decrease in magnitude becomes most pronounced near the normal incidence.
- At near the normal and grazing incidence angles, the magnitude of E_{int} decreases with the increase of pyro-electric parameters. However, in the mid-range of incidence angles, an opposite trend is observed.

The variations of the energy ratios for different values of frequency $\omega = 450$ Hz, 550 Hz, 650 Hz and 750 Hz are shown in Figures 6(a)–6(f). The important findings are as follows:

- Increasing the frequency leads to an increase in the magnitudes of E_{RP}, E_{RM}. This increase in magnitude becomes more noticeable as the angle of incidence increases $0 < \theta_0 \leq 90°$.
- Initially, the magnitudes of E_{RT} and E_{int} are not significantly affected by the frequency. However, after a certain angle $\theta_0 > 35°$, the influence of frequency becomes apparent and directly proportional to E_{RT} and E_{int}.
- Closer examination of the magnified sub-indent picture confirms that the magnitudes of E_{RS} and E_{REP} decreases significantly as the frequency is increased in the range of angle of incidence $0 < \theta_0 \leq 52°$ and $0 < \theta_0 \leq 22°$, respectively.

10. Conclusions

The reflection behavior of PHTE waves in the orthotropic medium is investigated by using the normal mode technique. The amplitude ratios were used to obtain the energy ratios of various reflected waves for an incident QP wave. The following conclusions may be made in light:

- The energy ratios of all reflected waves drop with increasing moisture content and pyro-electricity. This impact is more pronounced at higher incidence angles.
- At grazing incidence, the energy ratios of all reflected waves except E_{RM} decrease with increasing moisture diffusivity.
- The energy ratios of all reflected waves except E_{RP}, E_{RM} near the normal incidence are sufficiently affected by temperature diffusivity.
- At higher incidence angles, the energy ratios of all reflected waves increase with increasing frequency.
- These results might potentially affect the design of innovative materials and composites with specialized moisture, frequency and pyro-electric characteristics for improved energy storage and conversion.

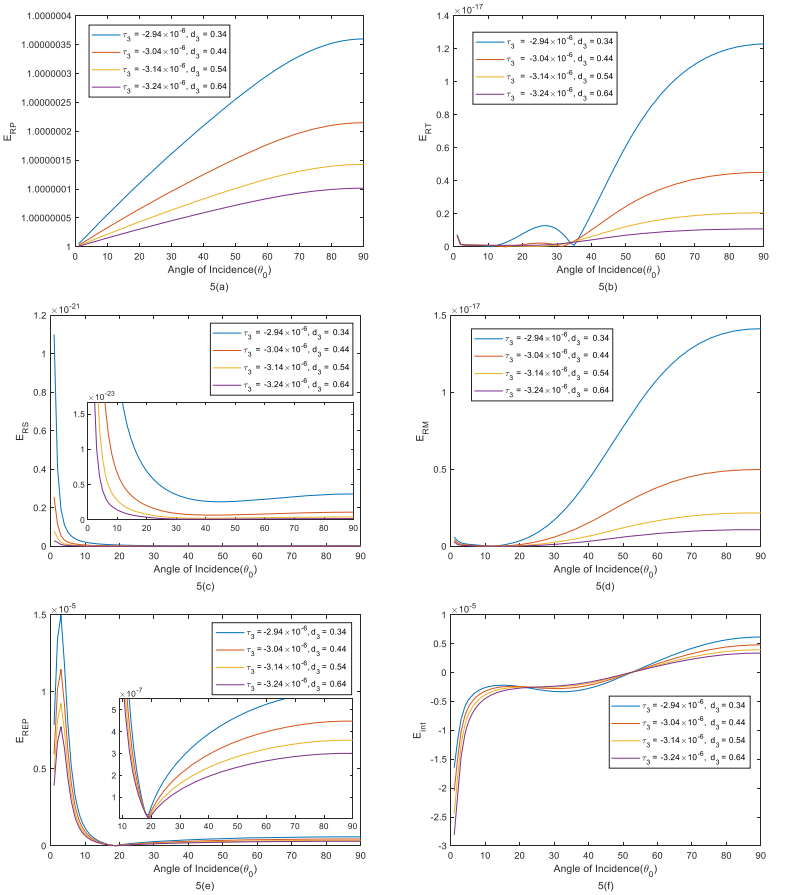
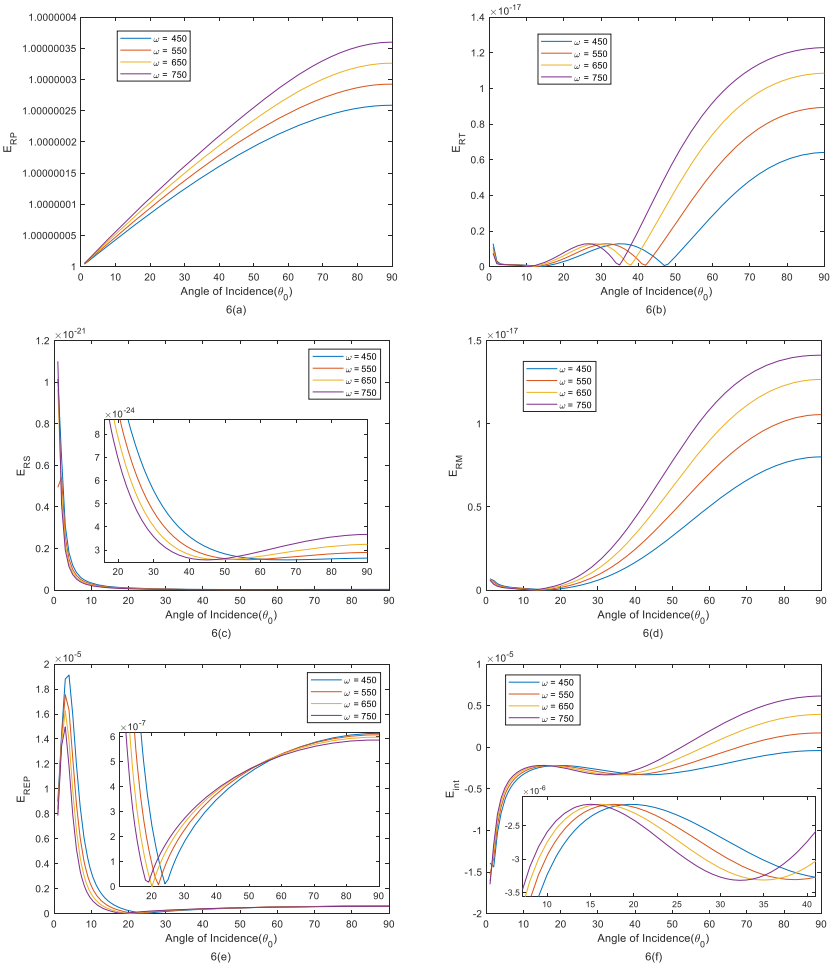


Figure 5.
Figures 5(a)–5(f). Variations of the energy ratios of reflected quasi-longitudinal P wave (E_{RP}), quasi-thermal wave (E_{RT}), quasi-transverse wave (E_{RS}), quasi-moisture wave (E_{RM}), electric potential wave (E_{REP}) and interaction energy (E_{int}) are depicted against the angle of incidence θ_0, $0 < \theta_0 < 90°$ for different values of the pyro-electric parameter ($\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$), ($\tau_3 = -3.04 \times 10^{-6}$, $d_3 = 0.44$), ($\tau_3 = -3.14 \times 10^{-6}$, $d_3 = 0.54$), ($\tau_3 = -3.24 \times 10^{-6}$, $d_3 = 0.64$), at when $m_0 = 10\%$, $D_1^T = 50$, $D_3^T = 75$, $D_1^m = 0.8 \times 10^{-4}$, $D_3^m = 1.1 \times 10^{-4}$, $\tau_3 = -2.94 \times 10^{-6}$, $d_3 = 0.34$ and $\omega = 750$ Hz

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Figure 6.
Figures 6(a)–6(f). Variations of the energy ratios of reflected quasi-longitudinal P wave (E_{RP}), quasi-thermal wave (E_{RT}), quasi-transverse wave (E_{RS}), quasi-moisture wave (E_{RM}), electric potential wave (E_{REP}) and interaction energy (E_{int}) are depicted, against the angle of incidence θ_0, $0 < \theta_0 < 90°$ for different values of frequency $\omega = 450$ Hz, 550 Hz, 650 Hz and 750 Hz, at when $m_0 = 10\%$, $D_1^T = 50$, $D_3^T = 75$, $D_1^m = 0.8 \times 10^{-4}$, $D_3^m = 1.1 \times 10^{-4}$, $\tau_3 = -2.94 \times 10^{-6}$ and $d_3 = 0.34$

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Appendix, compiled and presented by authors

$$B_0 = Q_3^T c_{55}\eta_{33}^2 q_3^m - D_3^T D_3^m c_{55}\eta_{33}^2 - D_3^T D_3^m c_{33}c_{55}\varepsilon_{33} + Q_3^T c_{33}c_{55}\varepsilon_{33}q_3^m$$

$$\begin{aligned}
B_1 = {} & D_3^T a_{45}c_{55}\eta_{33}^2 - D_3^T D_3^m a_{11}\eta_{33}^2 + D_3^m a_{34}c_{55}\eta_{33}^2 - Q_3^T a_{35}c_{55}\eta_{33}^2 + Q_3^T a_{11}\eta_{33}^2 q_3^m \\
& - a_{44}c_{55}\eta_{33}^2 q_3^m - D_3^T D_3^m a_{13}\, a_{51}c_{33} + D_3^T D_3^m a_{53}c_{33}c_{55} + D_3^T D_3^m a_{12}a_{21}\varepsilon_{33} \\
& + D_3^T D_3^m a_{13}a_{21}\eta_{33} + D_3^T D_3^m a_{12}a_{51}\eta_{33} - D_3^T D_3^m a_{11}c_{33}\,\varepsilon_{33} + D_3^T D_3^m a_{22}c_{55}\varepsilon_{33} \\
& + D_3^T D_3^m a_{23}c_{55}\eta_{33} + D_3^T D_3^m a_{52}c_{55}\eta_{33} - D_3^T a_{43}c_{33}c_{55}d_3 + D_3^m a_{32}\beta_3^T c_{55}\varepsilon_{33} \\
& + D_3^T a_{42}\beta_3^m c_{55}\varepsilon_{33} - D_3^m a_{33}\beta_3^T c_{55}\eta_{33} - D_3^T a_{43}\beta_3^m c_{55}\eta_{33} + D_3^T a_{45}c_{33}c_{55}\varepsilon_{33} \\
& + D_3^m a_{34}c_{33}c_{55}\varepsilon_{33} - D_3^T a_{42}c_{55}d_3\eta_{33} + Q_3^T a_{33}c_{33}c_{55}d_3 - Q_3^T a_{32}\beta_3^m c_{55}\varepsilon_{33} \\
& + Q_3^T a_{33}\beta_3^m c_{55}\eta_{33} - Q_3^T a_{35}c_{33}c_{55}\varepsilon_{33} + Q_3^T a_{32}c_{55}\, d_3\eta_{33} - D_3^m a_{33}c_{33}c_{55}\tau_3 \\
& - D_3^m a_{32}c_{55}\eta_{33}\tau_3 + Q_3^T a_{13}a_{51}c_{33}q_3^m - Q_3^T a_{53}c_{33}c_{55}q_3^m - Q_3^T a_{12}a_{21}\varepsilon_{33}q_3^m \\
& - Q_3^T a_{13}\, a_{21}\eta_{33}q_3^m - Q_3^T a_{12}a_{51}\eta_{33}q_3^m + Q_3^T a_{11}c_{33}\varepsilon_{33}q_3^m - Q_3^T a_{22}c_{55}\varepsilon_{33}q_3^m \\
& - Q_3^T a_{23}c_{55}\eta_{33}q_3^m - Q_3^T a_{52}c_{55}\eta_{33}q_3^m - a_{42}\beta_3^T c_{55}\varepsilon_{33}q_3^m + a_{43}\beta_3^T c_{55}\eta_{33}q_3^m \\
& - a_{44}c_{33}c_{55}\varepsilon_{33}q_3^m + a_{43}c_{33}c_{55}q_3^m\tau_3 + a_{42}c_{55}\eta_{33}q_3^m\tau_3
\end{aligned}$$

$$
\begin{aligned}
B_2 = {} & D_3^T a_{11} a_{45} \eta_{33}^2 - D_3^T a_{15} a_{41} \eta_{33}^2 + D_3^m a_{11} a_{34} \eta_{33}^2 - D_3^m a_{14} a_{31} \eta_{33}^2 - Q_3^T a_{11} a_{35} \eta_{33}^2 \\
& + Q_3^T a_{15} a_{31} \eta_{33}^2 - a_{34} a_{45}\, c_{55} \eta_{33}^2 + a_{35} a_{44} c_{55} \eta_{33}^2 - a_{11} a_{44} \eta_{33}^2 q_3^m + D_3^m a_{14} a_{31} \eta_{33}^2 \\
& - Q_3^T a_{11} a_{35} \eta_{33}^2 + Q_3^T a_{15} a_{31} \eta_{33}^2 - a_{34} a_{45} c_{55} \eta_{33}^2 + a_{35} a_{44} c_{55} \eta_{33}^2 - a_{11} a_{44} \eta_{33}^2 q_3^m \\
& + a_{14} a_{41} \eta_{33}^2 q_3^m - D_3^T D_3^m a_{12} a_{21} a_{53} - D_3^T D_3^m a_{12} a_{23} a_{51} - D_3^T D_3^m a_{13} a_{21}\, a_{52} \\
& + D_3^T D_3^m a_{13} a_{22} a_{51} + D_3^T D_3^m a_{11} a_{53} c_{33} - D_3^T D_3^m a_{22} a_{53} c_{55} - D_3^T D_3^m a_{23} a_{52} c_{55} \\
& + D_3^T D_3^m a_{11} a_{22} \varepsilon_{33} + D_3^T D_3^m a_{11} a_{23} \eta_{33} + D_3^T D_3^m a_{11} a_{52} \eta_{33} + D_3^m a_{12} a_{33} a_{51} \beta_3^T \\
& + D_3^m a_{13} a_{32} a_{51} \beta_3^T + D_3^T a_{12} a_{43} a_{51} \beta_3^m + D_3^T a_{13} a_{42} a_{51} \beta_3^m + D_3^T a_{13} a_{45} a_{51} c_{33} \\
& - D_3^T a_{15} a_{43} a_{51} c_{33} + D_3^m a_{13} a_{34} a_{51} c_{33} - D_3^m a_{14} a_{33} a_{51} c_{33} + D_3^T a_{12} a_{21} a_{43} d_3 \\
& + D_3^T a_{13} a_{21} a_{42} d_3 - D_3^m a_{32} a_{53} \beta_3^T c_{55} + D_3^m a_{33} a_{52} \beta_3^T c_{55} - D_3^T a_{42} a_{53} \beta_3^m c_{55} \\
& + D_3^T a_{43} a_{52} \beta_3^m c_{55} - D_3^T a_{45} a_{53} c_{33} c_{55} - D_3^m a_{34} a_{53} c_{33} c_{55} - D_3^T a_{12} a_{21} a_{45} \varepsilon_{33} \\
& - D_3^T a_{15} a_{21} a_{42} \varepsilon_{33} - D_3^m a_{12} a_{21}\, a_{34} \varepsilon_{33} - D_3^m a_{14} a_{21} a_{32} \varepsilon_{33} - D_3^T a_{13} a_{21} a_{45} \eta_{33} \\
& + D_3^T a_{15} a_{21} a_{43} \eta_{33} - D_3^m a_{13} a_{21} a_{34} \eta_{33} + D_3^m a_{14} a_{21} a_{33} \eta_{33} - D_3^T a_{12} a_{45} a_{51} \eta_{33} \\
& - D_3^T a_{15} a_{42} a_{51} \eta_{33} - D_3^m a_{12} a_{34} a_{51} \eta_{33} - D_3^m a_{14} a_{32} a_{51} \eta_{33} \\
& - D_3^T a_{11} a_{43} c_{33} d_3 D_3^T a_{13} a_{41} c_{33} d_3 + D_3^T a_{22} a_{43} c_{55} d_3 + D_3^T a_{23} a_{42} c_{55} d_3 + D_3^m a_{11} a_{32} \beta_3^T \varepsilon_{33} \\
& + D_3^m a_{12} a_{31} \beta_3^T \varepsilon_{33} + D_3^T a_{11} a_{42} \beta_3^m \varepsilon_{33} + D_3^T a_{12} a_{41} \beta_3^m \varepsilon_{33} - D_3^m a_{11} a_{33} \beta_3^T \eta_{33} \\
& + D_3^m a_{13} a_{31} \beta_3^T \eta_{33} - D_3^T a_{11} a_{43} \beta_3^m \eta_{33} + D_3^T a_{13} a_{41} \beta_3^m \eta_{33} + D_3^T a_{11} a_{45} c_{33} \varepsilon_{33} \\
& - D_3^T a_{15} a_{41} c_{33} \varepsilon_{33} + D_3^m a_{11} a_{34} c_{33} \varepsilon_{33} - D_3^m a_{14} a_{31} c_{33} \varepsilon_{33} - D_3^T a_{22} a_{45} c_{55} \varepsilon_{33} \\
& - D_3^m a_{22} a_{34} c_{55} \varepsilon_{33} - D_3^T a_{23} a_{45} c_{55} \eta_{33} - D_3^m a_{23} a_{34} c_{55} \eta_{33} - D_3^T a_{45} a_{52} c_{55} \eta_{33} \\
& - D_3^m a_{34} a_{52} c_{55} \eta_{33} - D_3^T a_{11} a_{42} d_3 \eta_{33} - D_3^T a_{12} a_{41} d_3 \eta_{33} - Q_3^T a_{12} a_{33} a_{51} \beta_3^m \\
& - Q_3^T a_{13} a_{32} a_{51} \beta_3^m - Q_3^T a_{13} a_{35} a_{51} c_{33} + Q_3^T a_{15} a_{33} a_{51} c_{33} - Q_3^T a_{12} a_{21} a_{33} d_3 \\
& - Q_3^T a_{13} a_{21} a_{32} d_3 + Q_3^T a_{32} a_{53} \beta_3^m c_{55} - Q_3^T a_{33} a_{52} \beta_3^m c_{55} + Q_3^T a_{35} a_{53} c_{33} c_{55} \\
& + Q_3^T a_{12} a_{21} a_{35} \varepsilon_{33} + Q_3^T a_{15} a_{21} a_{32} \varepsilon_{33} + Q_3^T a_{13} a_{21} a_{35} \eta_{33} - Q_3^T a_{15} a_{21} a_{33} \eta_{33} \\
& + Q_3^T a_{12} a_{35} a_{51} \eta_{33} + Q_3^T a_{15} a_{32} a_{51} \eta_{33} + Q_3^T a_{11} a_{33} c_{33} d_3 - Q_3^T a_{13} a_{31} c_{33} d_3 \\
& - Q_3^T a_{22} a_{33} c_{55} d_3 - Q_3^T a_{23} a_{32} c_{55} d_3 - Q_3^T a_{11} a_{32} \beta_3^m \varepsilon_{33} - Q_3^T a_{12} a_{31} \beta_3^m \varepsilon_{33} \\
& + Q_3^T a_{11} a_{33} \beta_3^m \eta_{33} - Q_3^T a_{13} a_{31} \beta_3^m \eta_{33} + D_3^m a_{12} a_{21} a_{33} \tau_3 + D_3^m a_{13} a_{21} a_{32} \tau_3
\end{aligned}
$$

$$
\begin{aligned}
&-Q_3^T a_{11}a_{35}c_{33}\varepsilon_{33} + Q_3^T a_{15}a_{31}c_{33}\varepsilon_{33} + Q_3^T a_{22}a_{35}c_{55}\varepsilon_{33} + Q_3^T a_{23}a_{35}c_{55}\eta_{33} + Q_3^T a_{35}a_{52}c_{55}\eta_{33}\\
&+ Q_3^T a_{11}a_{32}d_3\eta_{33} + Q_3^T a_{12}a_{31}d_3\eta_{33} - D_3^m a_{11}a_{33}c_{33}\tau_3 + D_3^m a_{13}a_{31}c_{33}\tau_3 + D_3^m a_{22}a_{33}c_{55}\tau_3\\
&+ D_3^m a_{23}a_{32}c_{55}\tau_3 - D_3^m a_{11}a_{32}\eta_{33}\tau_3 - D_3^m a_{12}a_{31}\eta_{33}\tau_3 + Q_3^T a_{12}a_{21}a_{53}q_3^m + Q_3^T a_{12}a_{23}a_{51}q_3^m\\
&+ Q_3^T a_{13}a_{21}a_{52}q_3^m - Q_3^T a_{13}a_{22}a_{51}q_3^m - Q_3^T a_{11}a_{53}c_{33}q_3^m + Q_3^T a_{22}a_{53}c_{55}q_3^m\\
&+ Q_3^T a_{23}a_{52}c_{55}q_3^m - Q_3^T a_{11}a_{22}\varepsilon_{33}q_3^m - Q_3^T a_{11}a_{23}\eta_{33}q_3^m - Q_3^T a_{11}a_{52}\eta_{33}q_3^m\\
&+ a_{32}a_{43}\beta_3^T c_{55}d_3 - a_{33}a_{42}\beta_3^T c_{55}d_3 - a_{33}a_{44}c_{33}c_{55}d_3 + a_{34}a_{43}c_{33}c_{55}d_3 - a_{32}a_{45}\beta_3^T c_{55}\varepsilon_{33}\\
&+ a_{35}a_{42}\beta_3^T c_{55}\varepsilon_{33} + a_{32}a_{44}\beta_3^m c_{55}\varepsilon_{33} - a_{34}a_{42}\beta_3^m c_{55}\varepsilon_{33} + a_{33}a_{45}\beta_3^T c_{55}\eta_{33}\\
&- a_{35}a_{43}\beta_3^T c_{55}\eta_{33} - a_{33}a_{44}\beta_3^m c_{55}\eta_{33} + a_{34}a_{43}\beta_3^m c_{55}\eta_{33} - a_{34}a_{45}c_{33}c_{55}\varepsilon_{33}\\
&+ a_{35}a_{44}c_{33}c_{55}\varepsilon_{33} - a_{32}a_{44}c_{55}d_3\eta_{33} + a_{34}a_{42}c_{55}d_3\eta_{33} - a_{12}a_{43}a_{51}\beta_3^T q_3^m - a_{13}a_{42}a_{51}\beta_3^T q_3^m\\
&- a_{13}a_{44}a_{51}c_{33}q_3^m + a_{14}a_{43}a_{51}c_{33}q_3^m + a_{42}a_{53}\beta_3^T c_{55}q_3^m - a_{43}a_{52}\beta_3^T c_{55}q_3^m + a_{44}a_{53}c_{33}c_{55}q_3^m\\
&+ a_{12}a_{21}a_{44}\varepsilon_{33}q_3^m + a_{14}a_{21}a_{42}\varepsilon_{33}q_3^m + a_{13}a_{21}a_{44}\eta_{33}q_3^m - a_{14}a_{21}a_{43}\eta_{33}q_3^m\\
&+ a_{12}a_{44}a_{51}\eta_{33}q_3^m + a_{14}a_{42}a_{51}\eta_{33}q_3^m - a_{11}a_{42}\beta_3^T\eta_{33}q_3^m - a_{12}a_{41}\beta_3^T\varepsilon_{33}q_3^m\\
&+ a_{11}a_{43}\beta_3^T\eta_{33}q_3^m - a_{13}a_{41}\beta_3^T\eta_{33}q_3^m - a_{11}a_{44}c_{33}\varepsilon_{33}q_3^m + a_{14}a_{41}c_{33}\varepsilon_{33}q_3^m + a_{22}a_{44}c_{55}\varepsilon_{33}q_3^m\\
&+ a_{23}a_{44}c_{55}\eta_{33}q_3^m + a_{44}a_{52}c_{55}\eta_{33}q_3^m - a_{32}a_{43}\beta_3^m c_{55}\tau_3 + a_{33}a_{42}\beta_3^m c_{55}\tau_3 + a_{33}a_{45}c_{33}c_{55}\tau_3\\
&- a_{35}a_{43}c_{33}c_{55}\tau_3 + a_{32}a_{45}c_{55}\eta_{33}\tau_3 - a_{35}a_{42}c_{55}\eta_{33}\tau_3 - a_{12}a_{21}a_{43}q_3^m\tau_3 - a_{13}a_{21}a_{42}q_3^m\tau_3\\
&+ a_{11}a_{43}c_{33}q_3^m\tau_3 - a_{13}a_{41}c_{33}q_3^m\tau_3 - a_{22}a_{43}c_{55}q_3^m\tau_3 - a_{23}a_{42}c_{55}q_3^m\tau_3 + a_{11}a_{42}\eta_{33}q_3^m\tau_3\\
&+ a_{12}a_{41}\eta_{33}q_3^m\tau_3
\end{aligned}
$$

$$
\begin{aligned}
B_3 = {} & (-a_{11}a_{22}a_{53} - a_{11}a_{23}a_{52})D_3^T D_3^m \\
& + (a_{15}a_{23}a_{41} - a_{11}a_{23}a_{45} - a_{11}a_{45}a_{52} + a_{15}a_{41}a_{52})D_3^T \eta_{33} \\
& + (a_{15}a_{22}a_{41} - a_{11}a_{22}a_{45})D_3^T \varepsilon_{33} \\
& + (a_{11}a_{43}a_{52} - a_{11}a_{42}a_{53} - a_{12}a_{41}a_{53} - a_{13}a_{41}a_{52})D_3^T \beta_3^m \\
& + (a_{12}a_{21}a_{45}a_{53} + a_{12}a_{23}a_{45}a_{51} + a_{13}a_{21}a_{45}a_{52} - a_{13}a_{22}a_{45}a_{51} + a_{15}a_{21}a_{42}a_{53} \\
& - a_{15}a_{21}a_{43}a_{52} + a_{15}a_{22}a_{43}a_{51} + a_{15}a_{23}a_{42}a_{51} - a_{11}a_{45}a_{53}c_{33} + a_{15}a_{41}a_{53}c_{33} \\
& + a_{22}a_{45}a_{53}c_{55} + a_{23}a_{45}a_{52}c_{55} + a_{11}a_{22}a_{43}d_3 + a_{11}a_{23}a_{42}d_3 + a_{12}a_{23}a_{41}d_3 \\
& - a_{13}a_{22}a_{41}d_3)D_3^T + (a_{14}a_{23}a_{31} - a_{11}a_{23}a_{34} - a_{11}a_{34}a_{52} + a_{14}a_{31}a_{52})D_3^m \eta_{33} \\
& + (a_{14}a_{22}a_{31} - a_{11}a_{22}a_{34})D_3^m \varepsilon_{33} \\
& + (a_{11}a_{33}a_{52} - a_{11}a_{32}a_{53} - a_{12}a_{31}a_{53} - a_{13}a_{31}a_{52})D_3^m \beta_3^T \\
& + (a_{11}a_{22}a_{33} + a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} - a_{13}a_{22}a_{31})D_3^m \tau_3 \\
& + (a_{12}a_{21}a_{34}a_{53} + a_{12}a_{23}a_{34}a_{51} + a_{13}a_{21}a_{34}a_{52} - a_{13}a_{22}a_{34}a_{51} + a_{14}a_{21}a_{32}a_{53} \\
& - a_{14}a_{21}a_{33}a_{52} + a_{14}a_{22}a_{33}a_{51} + a_{14}a_{23}a_{32}a_{51} - a_{11}a_{34}a_{53}c_{33} + a_{14}a_{31}a_{53}c_{33} \\
& + a_{22}a_{34}a_{53}c_{55} + a_{23}a_{34}a_{52}c_{55})D_3^m \\
& + (a_{11}a_{35}a_{44} - a_{11}a_{34}a_{45} + a_{14}a_{31}a_{45} - a_{14}a_{35}a_{41} - a_{15}a_{31}a_{44} + a_{15}a_{34}a_{41})\eta_{33}^2 \\
& + (a_{11}a_{34}a_{43} - a_{11}a_{33}a_{44} + a_{13}a_{31}a_{44} - a_{13}a_{34}a_{41} - a_{14}a_{31}a_{43} + a_{14}a_{33}a_{41})\eta_{33}\beta_3^m \\
& + (a_{11}a_{33}a_{45} - a_{11}a_{35}a_{43} - a_{13}a_{31}a_{45} + a_{13}a_{35}a_{41} + a_{15}a_{31}a_{43} - a_{15}a_{33}a_{41})\eta_{33}\beta_3^T \\
& + (a_{11}a_{32}a_{45} - a_{11}a_{35}a_{42} + a_{12}a_{31}a_{45} - a_{12}a_{35}a_{41} + a_{15}a_{31}a_{42} - a_{15}a_{32}a_{41})\eta_{33}\tau_3 \\
& + (Q_3^T a_{11}a_{23}a_{35} - Q_3^T a_{15}a_{23}a_{31} + Q_3^T a_{11}a_{35}a_{52} - Q_3^T a_{15}a_{31}a_{52} + a_{13}a_{21}a_{34}a_{45} \\
& - a_{13}a_{21}a_{35}a_{44} - a_{14}a_{21}a_{33}a_{45} + a_{14}a_{21}a_{35}a_{43} + a_{15}a_{21}a_{33}a_{44} - a_{15}a_{21}a_{34}a_{43} \\
& + a_{12}a_{34}a_{45}a_{51} - a_{12}a_{35}a_{44}a_{51} + a_{14}a_{32}a_{45}a_{51} - a_{14}a_{35}a_{42}a_{51} - a_{15}a_{32}a_{44}a_{51} \\
& + a_{15}a_{34}a_{42}a_{51} + a_{23}a_{34}a_{45}c_{55} - a_{23}a_{35}a_{44}c_{55} + a_{34}a_{45}a_{52}c_{55} - a_{35}a_{44}a_{52}c_{55} \\
& - a_{11}a_{32}a_{44}d_3 + a_{11}a_{34}a_{42}d_3 - a_{12}a_{31}a_{44}d_3 + a_{12}a_{34}a_{41}d_3 - a_{14}a_{31}a_{42}d_3 + a_{14}a_{32}a_{41}d_3 \\
& + a_{11}a_{23}a_{44}q_3^m - a_{14}a_{23}a_{41}q_3^m + a_{11}a_{44}a_{52}q_3^m - a_{14}a_{41}a_{52}q_3^m)\eta_{33} \\
& + (a_{11}a_{32}a_{44} - a_{11}a_{34}a_{42} + a_{12}a_{31}a_{44} - a_{12}a_{34}a_{41} + a_{14}a_{31}a_{42} - a_{14}a_{32}a_{41})\varepsilon_{33}\beta_3^m \\
& + (a_{11}a_{35}a_{42} - a_{11}a_{32}a_{45} - a_{12}a_{31}a_{45} + a_{12}a_{35}a_{41} - a_{15}a_{31}a_{42} + a_{15}a_{32}a_{41})\varepsilon_{33}\beta_3^T
\end{aligned}
$$

$$
\begin{aligned}
&+(Q_3^T a_{11}a_{22}a_{35} - Q_3^T a_{15}a_{22}a_{31} + a_{22}a_{34}a_{45}c_{55} - a_{22}a_{35}a_{44}c_{55} + a_{11}a_{22}a_{44}q_3^m \\
&- a_{14}a_{22}a_{41}q_3^m)\varepsilon_{33} \\
&+(a_{11}a_{33}a_{42} - a_{11}a_{32}a_{43} - a_{12}a_{31}a_{43} + a_{12}a_{33}a_{41} - a_{13}a_{31}a_{42} + a_{13}a_{32}a_{41})\beta_3^m\tau_3 \\
&+(Q_3^T a_{11}a_{32}a_{53} - Q_3^T a_{11}a_{33}a_{52} + Q_3^T a_{12}a_{31}a_{53} + Q_3^T a_{13}a_{31}a_{52} + a_{12}a_{33}a_{44}a_{51} \\
&- a_{12}a_{34}a_{43}a_{51} + a_{13}a_{32}a_{44}a_{51} - a_{13}a_{34}a_{42}a_{51} - a_{14}a_{32}a_{43}a_{51} + a_{14}a_{33}a_{42}a_{51} \\
&- a_{32}a_{44}a_{53}c_{55} + a_{33}a_{44}a_{52}c_{55} + a_{34}a_{42}a_{53}c_{55} - a_{34}a_{43}a_{52}c_{55})\beta_3^m \\
&+(a_{12}a_{35}a_{43}a_{51} - a_{12}a_{33}a_{45}a_{51} - a_{13}a_{32}a_{45}a_{51} + a_{13}a_{35}a_{42}a_{51} + a_{15}a_{32}a_{43}a_{51} \\
&- a_{15}a_{33}a_{42}a_{51} + a_{32}a_{45}a_{53}c_{55} - a_{33}a_{45}a_{52}c_{55} - a_{35}a_{42}a_{53}c_{55} + a_{13}a_{35}a_{42}a_{51} \\
&+ a_{15}a_{32}a_{43}a_{51} - a_{15}a_{33}a_{42}a_{51} + a_{32}a_{45}a_{53}c_{55} - a_{33}a_{45}a_{52}c_{55} - a_{35}a_{42}a_{53}c_{55} \\
&+ a_{35}a_{43}a_{52}c_{55} + a_{11}a_{32}a_{43}d_3 - a_{11}a_{33}a_{42}d_3 + a_{12}a_{31}a_{43}d_3 - a_{12}a_{33}a_{41}d_3 + a_{13}a_{31}a_{42}d_3 \\
&- a_{13}a_{32}a_{41}d_3 + a_{11}a_{42}a_{53}q_3^m - a_{11}a_{43}a_{52}q_3^m + a_{12}a_{41}a_{53}q_3^m + a_{13}a_{41}a_{52}q_3^m)\beta_3^T \\
&+(a_{12}a_{21}a_{35}a_{43} - a_{12}a_{21}a_{33}a_{45} - a_{13}a_{21}a_{32}a_{45} + a_{13}a_{21}a_{35}a_{42} + a_{15}a_{21}a_{32}a_{43} \\
&- a_{15}a_{21}a_{33}a_{42} + a_{11}a_{33}a_{45}c_{33} - a_{11}a_{35}a_{43}c_{33} - a_{13}a_{31}a_{45}c_{33} + a_{13}a_{35}a_{41}c_{33} + a_{15}a_{31}a_{43}c_{33} \\
&- a_{15}a_{33}a_{41}c_{33} - a_{22}a_{33}a_{45}c_{55} + a_{22}a_{35}a_{43}c_{55} - a_{23}a_{32}a_{45}c_{55} + a_{23}a_{35}a_{42}c_{55} - a_{11}a_{22}a_{43}q_3^m \\
&- a_{11}a_{23}a_{42}q_3^m - a_{12}a_{23}a_{41}q_3^m + a_{13}a_{22}a_{41}q_3^m)\tau_3 + Q_3^T a_{13}a_{22}a_{35}a_{51} - Q_3^T a_{12}a_{23}a_{35}a_{51} \\
&- Q_3^T a_{13}a_{21}a_{35}a_{52} - Q_3^T a_{12}a_{21}a_{35}a_{53} - Q_3^T a_{15}a_{21}a_{32}a_{53} + Q_3^T a_{15}a_{21}a_{33}a_{52} \\
&- Q_3^T a_{15}a_{22}a_{33}a_{51} - Q_3^T a_{15}a_{23}a_{32}a_{51} + Q_3^T a_{11}a_{35}a_{53}c_{33} - Q_3^T a_{15}a_{31}a_{53}c_{33} \\
&- Q_3^T a_{22}a_{35}a_{53}c_{55} - Q_3^T a_{23}a_{35}a_{52}c_{55} - Q_3^T a_{11}a_{22}a_{33}d_3 - Q_3^T a_{11}a_{23}a_{32}d_3 - Q_3^T a_{12}a_{23}a_{31}d_3 \\
&+ Q_3^T a_{13}a_{22}a_{31}d_3 + Q_3^T a_{11}a_{22}a_{53}q_3^m + Q_3^T a_{11}a_{23}a_{52}q_3^m - a_{13}a_{34}a_{45}a_{51}c_{33} + a_{13}a_{35}a_{44}a_{51}c_{33} \\
&+ a_{14}a_{33}a_{45}a_{51}c_{33} - a_{14}a_{35}a_{43}a_{51}c_{33} - a_{15}a_{33}a_{44}a_{51}c_{33} + a_{15}a_{34}a_{43}a_{51}c_{33} + a_{12}a_{21}a_{33}a_{44}d_3 \\
&- a_{12}a_{21}a_{34}a_{43}d_3 + a_{13}a_{21}a_{32}a_{44}d_3 - a_{13}a_{21}a_{34}a_{42}d_3 - a_{14}a_{21}a_{32}a_{43}d_3 + a_{14}a_{21}a_{33}a_{42}d_3 \\
&+ a_{34}a_{45}a_{53}c_{33}c_{55} - a_{35}a_{44}a_{53}c_{33}c_{55} - a_{11}a_{33}a_{44}c_{33}d_3 + a_{11}a_{34}a_{43}c_{33}d_3 + a_{13}a_{31}a_{44}c_{33}d_3 \\
&- a_{13}a_{34}a_{41}c_{33}d_3 - a_{14}a_{31}a_{43}c_{33}d_3 + a_{14}a_{33}a_{41}c_{33}d_3 + a_{22}a_{33}a_{44}c_{55}d_3 - a_{22}a_{34}a_{43}c_{55}d_3 \\
&+ a_{23}a_{32}a_{44}c_{55}d_3 - a_{23}a_{34}a_{42}c_{55}d_3 \\
&+(-a_{12}a_{21}a_{44}a_{53} - a_{12}a_{23}a_{44}a_{51} - a_{13}a_{21}a_{44}a_{52} + a_{13}a_{22}a_{44}a_{51} - a_{14}a_{21}a_{42}a_{53} \\
&+ a_{14}a_{21}a_{43}a_{52} - a_{14}a_{22}a_{43}a_{51} - a_{14}a_{23}a_{42}a_{51} + a_{11}a_{44}a_{53}c_{33} - a_{14}a_{41}a_{53}c_{33} \\
&- a_{22}a_{44}a_{53}c_{55} - a_{23}a_{44}a_{52}c_{55})q_3^m
\end{aligned}
$$

$$
\begin{aligned}
B_4 = {} & (a_{11}a_{22}a_{45}a_{53} + a_{11}a_{23}a_{45}a_{52} - a_{15}a_{22}a_{41}a_{53} - a_{15}a_{23}a_{41}a_{52})D_3^T \\
& + (a_{11}a_{22}a_{34}a_{53} + a_{11}a_{23}a_{34}a_{52} - a_{14}a_{22}a_{31}a_{53} - a_{14}a_{23}a_{31}a_{52})D_3^m \\
& + (a_{11}a_{23}a_{34}a_{45} - a_{11}a_{23}a_{35}a_{44} - a_{14}a_{23}a_{31}a_{45} + a_{14}a_{23}a_{35}a_{41} + a_{15}a_{23}a_{31}a_{44} \\
& - a_{15}a_{23}a_{34}a_{41} + a_{11}a_{34}a_{45}a_{52} - a_{11}a_{35}a_{44}a_{52} - a_{14}a_{31}a_{45}a_{52} + a_{14}a_{35}a_{41}a_{52} \\
& + a_{15}a_{31}a_{44}a_{52} - a_{15}a_{34}a_{41}a_{52})\eta_{33} \\
& + (a_{11}a_{22}a_{34}a_{45} - a_{11}a_{22}a_{35}a_{44} - a_{14}a_{22}a_{31}a_{45} + a_{14}a_{22}a_{35}a_{41} + a_{15}a_{22}a_{31}a_{44} \\
& - a_{15}a_{22}a_{34}a_{41})\varepsilon_{33} \\
& + (a_{11}a_{33}a_{44}a_{52} - a_{11}a_{32}a_{44}a_{53} + a_{11}a_{34}a_{42}a_{53} - a_{11}a_{34}a_{43}a_{52} - a_{12}a_{31}a_{44}a_{53} \\
& + a_{12}a_{34}a_{41}a_{53} - a_{13}a_{31}a_{44}a_{52} + a_{13}a_{34}a_{41}a_{52} - a_{14}a_{31}a_{42}a_{53} + a_{14}a_{31}a_{43}a_{52} \\
& + a_{14}a_{32}a_{41}a_{53} - a_{14}a_{33}a_{41}a_{52})\beta_3^m \\
& + (a_{11}a_{32}a_{45}a_{53} - a_{11}a_{33}a_{45}a_{52} - a_{11}a_{35}a_{42}a_{53} + a_{11}a_{35}a_{43}a_{52} + a_{12}a_{31}a_{45}a_{53} \\
& - a_{12}a_{35}a_{41}a_{53} + a_{13}a_{31}a_{45}a_{52} - a_{13}a_{35}a_{41}a_{52} + a_{15}a_{31}a_{42}a_{53} - a_{15}a_{31}a_{43}a_{52} \\
& - a_{15}a_{32}a_{41}a_{53} + a_{15}a_{33}a_{41}a_{52})\beta_3^T \\
& + (a_{11}a_{22}a_{35}a_{43} - a_{11}a_{22}a_{33}a_{45} - a_{11}a_{23}a_{32}a_{45} + a_{11}a_{23}a_{35}a_{42} - a_{12}a_{23}a_{31}a_{45} \\
& + a_{12}a_{23}a_{35}a_{41} + a_{13}a_{22}a_{31}a_{45} - a_{13}a_{22}a_{35}a_{41} - a_{15}a_{22}a_{31}a_{43} + a_{15}a_{22}a_{33}a_{41} \\
& - a_{15}a_{23}a_{31}a_{42} + a_{15}a_{23}a_{32}a_{41})\tau_3 \\
& + (a_{15}a_{22}a_{31}a_{53} - a_{11}a_{23}a_{35}a_{52} - a_{11}a_{22}a_{35}a_{53} + a_{15}a_{23}a_{31}a_{52})Q_3^T - a_{12}a_{21}a_{34}a_{45}a_{53} \\
& + a_{12}a_{21}a_{35}a_{44}a_{53} - a_{12}a_{23}a_{34}a_{45}a_{51} + a_{12}a_{23}a_{35}a_{44}a_{51} - a_{13}a_{21}a_{34}a_{45}a_{52} \\
& + a_{13}a_{21}a_{35}a_{44}a_{52} + a_{13}a_{22}a_{34}a_{45}a_{51} - a_{13}a_{22}a_{35}a_{44}a_{51} - a_{14}a_{21}a_{32}a_{45}a_{53} \\
& + a_{14}a_{21}a_{33}a_{45}a_{52} + a_{14}a_{21}a_{35}a_{42}a_{53} - a_{14}a_{21}a_{35}a_{43}a_{52} - a_{14}a_{22}a_{33}a_{45}a_{51} \\
& + a_{14}a_{22}a_{35}a_{43}a_{51} - a_{14}a_{23}a_{32}a_{45}a_{51} + a_{14}a_{23}a_{35}a_{42}a_{51} + a_{15}a_{21}a_{32}a_{44}a_{53} \\
& - a_{15}a_{21}a_{33}a_{44}a_{52} - a_{15}a_{21}a_{34}a_{42}a_{53} + a_{15}a_{21}a_{34}a_{43}a_{52} + a_{15}a_{22}a_{33}a_{44}a_{51} \\
& - a_{15}a_{22}a_{34}a_{43}a_{51} + a_{15}a_{23}a_{32}a_{44}a_{51} - a_{15}a_{23}a_{34}a_{42}a_{51} + a_{11}a_{34}a_{45}a_{53}c_{33} \\
& - a_{11}a_{35}a_{44}a_{53}c_{33} - a_{14}a_{31}a_{45}a_{53}c_{33} + a_{14}a_{35}a_{41}a_{53}c_{33} + a_{15}a_{31}a_{44}a_{53}c_{33} \\
& - a_{15}a_{34}a_{41}a_{53}c_{33} - a_{22}a_{34}a_{45}a_{53}c_{55} + a_{22}a_{35}a_{44}a_{53}c_{55} - a_{23}a_{34}a_{45}a_{52}c_{55} \\
& + a_{23}a_{35}a_{44}a_{52}c_{55} + a_{11}a_{22}a_{33}a_{44}d_3 - a_{11}a_{22}a_{34}a_{43}d_3 + a_{11}a_{23}a_{32}a_{44}d_3 \\
& - a_{11}a_{23}a_{34}a_{42}d_3 + a_{12}a_{23}a_{31}a_{44}d_3 - a_{12}a_{23}a_{34}a_{41}d_3
\end{aligned}
$$

$$-a_{13}a_{22}a_{31}a_{44}d_3 + a_{13}a_{22}a_{34}a_{41}d_3 + a_{14}a_{22}a_{31}a_{43}d_3 - a_{14}a_{22}a_{33}a_{41}d_3 + a_{14}a_{23}a_{31}a_{42}d_3$$
$$- a_{14}a_{23}a_{32}a_{41}d_3 + (a_{14}a_{22}a_{41}a_{53} - a_{11}a_{22}a_{44}a_{53} - a_{11}a_{23}a_{44}a_{52} + a_{14}a_{23}a_{41}a_{52})q_3^m$$

$$B_5 = -a_{11}a_{22}a_{34}a_{45}a_{53} + a_{11}a_{22}a_{35}a_{44}a_{53} - a_{11}a_{23}a_{34}a_{45}a_{52} + a_{11}a_{23}a_{35}a_{44}a_{52}$$
$$+ a_{14}a_{22}a_{31}a_{45}a_{53} - a_{14}a_{22}a_{35}a_{41}a_{53} + a_{14}a_{23}a_{31}a_{45}a_{52} - a_{14}a_{23}a_{35}a_{41}a_{52}$$
$$- a_{15}a_{22}a_{31}a_{44}a_{53} + a_{15}a_{22}a_{34}a_{41}a_{53} - a_{15}a_{23}a_{31}a_{44}a_{52} + a_{15}a_{23}a_{34}a_{41}a_{52}$$

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